Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN | Date: 22-03-2026

Time: 09.00AM TO 12.00PM | GTM-26 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**3**	2	**1**	3	**2**	4	**2**	5	**4**
6	**2**	7	**4**	8	**3**	9	**2**	10	**3**
11	**3**	12	**1**	13	**3**	14	**4**	15	**2**
16	**4**	17	**2**	18	**1**	19	**3**	20	**2**
21	**19**	22	**384**	23	**270**	24	**180**	25	**8**

PHYSICS

26	**2**	27	**3**	28	**4**	29	**2**	30	**4**
31	**3**	32	**2**	33	**2**	34	**1**	35	**1**
36	**1**	37	**3**	38	**3**	39	**2**	40	**3**
41	**1**	42	**3**	43	**2**	44	**2**	45	**2**
46	**5**	47	**250**	48	**18**	49	**112**	50	**8**

CHEMISTRY

51	**1**	52	**3**	53	**3**	54	**2**	55	**4**
56	**1**	57	**3**	58	**1**	59	**3**	60	**1**
61	**4**	62	**1**	63	**3**	64	**1**	65	**4**
66	**1**	67	**1**	68	**1**	69	**2**	70	**4**
71	**3**	72	**12**	73	**2**	74	**2**	75	**3**

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SOLUTIONS

MATHEMATICS

01. $A=\begin{pmatrix}3 & 1\\ 7 & 5\end{pmatrix}$ and $A^2-yA+xI=0$

Characteristic equation is $A^2-\lambda_1 A+\lambda_2 I=0$

Where $\lambda_1=Tr(A)=8=y$

$\lambda_2=|A|=15-7=8=x$

Trace of $\begin{pmatrix}y & 2\\ 1 & x\end{pmatrix}$ is $y+x=16=2(x+y)=32$

02. Let $P(t_1^2 2t_1), Q(t_2^2, 2t_2)$ on parabola $y^2=4x$

Here slope of $\overline{PQ}=\dfrac{2t_2-2t_1}{t_2^2-t_1^2}=2\Rightarrow t_1+t_2=2$

Let $R(x_1,y_1)$ divides $\overline{PQ}$ in the ration 1:2

Now $x_1=\dfrac{2t_1^2+t_2^2}{3}$, and $y_1=\dfrac{4t_1+2t_2}{3}\Rightarrow 3x_1=2t_1^2+t_2^2, 3y_1=4t_1+2t_2$

$3x_1=2t_1^2+(1-t_1)^2, 3y_1=4t_1+2(1-t_2)\Rightarrow 3x_1=3t_1^2-2t_1+1,\ 3y_1=2t_1+2$

$\Rightarrow 3x_1=3\left(\dfrac{3y_1-2}{2}\right)^2-2\left(\dfrac{3y_1-2}{2}\right)+1\Rightarrow\left(y_1-\dfrac{8}{9}\right)^2=\dfrac{4}{9}\left(x_1-\dfrac{2}{9}\right)$

Locus is $\left(y-\dfrac{8}{9}\right)^2=\dfrac{4}{9}\left(x-\dfrac{2}{9}\right)$

Vertex is $(\alpha,\beta)=\left(\dfrac{2}{9},\dfrac{8}{9}\right)$

03. Area $=2\left(\int_0^1 x^2dx-\int_{3/4}^1\sqrt{4x-3}\,dx\right)=2\left(\dfrac{1}{3}-\dfrac{1}{6}\right)=\dfrac{1}{3}$

04.

$I_1=\int_0^{\pi}\dfrac{x\sin x}{1+\cos^2 x}dx=\dfrac{\pi}{2}\int_0^{\pi}\dfrac{\sin x}{1+\cos^2 x}dx$

$\pi\int_0^{\pi/2}\dfrac{\sin x}{1+\cos^2 x}dx$ put cos x= t

$\Rightarrow\pi\int_1^0\dfrac{-dt}{1+t^2}=-\pi\tan^{-1}t)_1^0=-\pi\left(0-\dfrac{\pi}{4}\right)=\dfrac{\pi^2}{4}$

$I_2=\int_0^{\pi}x\sin^4 xdx=\dfrac{\pi}{2}\int_0^{\pi}\sin^4 x\,dx=\pi\int_0^{\pi/2}\sin^4 xdx=\pi\times\dfrac{3}{4}\times\dfrac{1}{2}\times\dfrac{\pi}{2}=\dfrac{3\pi^2}{16}$

$I_1:2I_2=\dfrac{\pi^2}{4}:2\times\dfrac{3\pi^2}{16}=2:3$

05. No of diagonals $=\dfrac{n(n-3)}{2}=20\ \Rightarrow n=8$

a) $n_{c_3}=8_{c_3}=56$, b) $n_{c_4}=8_{c_4}=70$,c) $n(n-4)=8\times 4=32$,d) $n=8$

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06. Let $A(2,3), B(0,2), C(4,5), D(0,K)$

Equation circle passing through A, B is $x^2+y^2-5x+6+\lambda x-2y+4=0$

If passes through $C(4,5) \Rightarrow \lambda = 7$

$\therefore$ Equation of circle $x^2+y^2+5x+9y+34=0$

Sub D(0,K) in above equation

$\Rightarrow K^2-19K+34=0 \Rightarrow K=17,2 \Rightarrow$ Sum of possible values of K is 19

07. $\frac{4-x^2}{2+[x]} \geq 0$

Case I : $4-x^2 \geq 0, 2+[x]>0$

$\Rightarrow x \in [-2,2] and\ x \in [-1,\infty) \Rightarrow x \in [-1,2] \ldots\ldots\ldots\ldots\ldots\ldots (1)$

Case II: $4-x^2<0, 2+[x]<0 \Rightarrow x \in (-\infty,-2) \cup (2,\infty), x \in (-\infty,-2) \Rightarrow x \in (-\infty,-2) \ldots\ldots\ldots\ldots\ldots\ldots (2)$

$\therefore$ Domain is $(-\infty,-2) \cup [-1,2]$

08. $(m+n)p_2 = 90 \Rightarrow (m+n)(m+n-1)=90 \Rightarrow (m+n)^2-(m+n)-90=0 \Rightarrow m+n=10$

$(m-n)p_2 = 30 \Rightarrow (m-n)^2-(m-n)-30=0 \Rightarrow m-n=6$

No of divisors of $\lambda = m^2-n^2 = 2^2 \times 3^1 \times 5^1$ is $3 \times 2 \times 2 = 12$

09. Image A w.r. to $x-1=0$ is $P(-2,-1)$

Image of A w.r. to $x-y-1=0$ is $Q(0,3)$

Equation of $\overline{BC}$ is $2x-y+3=0$, solving $x-1=0$ and $2x-y+3=0$

We get $B(1,5)$

$\alpha = 1, \beta = 5$

10. $\bar{a}+\bar{b}+\bar{c}=\bar{0} \Rightarrow \bar{a}^2 = \bar{b}^2+\bar{c}^2+2\bar{b}.\bar{c} \Rightarrow 144 = 48+\bar{c} \Rightarrow \bar{c}^2 = 48 \Rightarrow |\bar{c}| = 4\sqrt{3}$

$\bar{c}^2 = \bar{a}^2+\bar{b}^2-2\bar{a}.\bar{b}$

$\bar{a}.\bar{b} = \frac{4.8-144-48}{2} = -72$

And $|\bar{a}\times\bar{b}+\bar{c}\times\bar{a}| = |\bar{a}\times\bar{b}+\bar{a}\times\bar{b}| = 2|\bar{a}\times\bar{b}| = 2\sqrt{\bar{a}^2\bar{b}^2-(\bar{a}-\bar{b})^2} = 48\sqrt{3}$

11. Z lies on the circle (or) inside the circle with centre C $(2,-1)$ and radius r=2

Greatest value of $|z|$ = CP+r

Least value of $|z|$ is $CP-r, where\ p=(0,0)$

Now $CP+r+CP-r = 2CP = 2\sqrt{5}$

12. $\sin^2 2\theta = 2\cos^2\theta$

$4\sin^2\theta\cos^2\theta = 2\cos^2\theta$

$\cos^2\theta (or) \sin^2\theta = \frac{1}{2} \Rightarrow \sin^2\theta = \pm\frac{1}{\sqrt{2}}$

$\frac{2\tan\theta}{1+\tan^2\theta} + \frac{1-\tan^2\theta}{1+\tan^2\theta} = \tan\theta \Rightarrow 2\tan\theta+1-\tan^2\theta = \tan\theta \Rightarrow \tan^3\theta+\tan^2\theta-\tan\theta-1=0$

$\tan\theta = 1-1$

$\therefore$ common solution is $\tan\theta = 1 \quad \therefore \Rightarrow \theta = \frac{-\pi}{4}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$

$\therefore$ no of values θ is 4

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13. $\int \frac{dx}{\sqrt{x}+3\sqrt{x}} =, put\ x = t^6$

$\int \frac{6t^5 dt}{t^3+t} = 6\int \frac{t^3}{t+1}dt \quad = 6\int \frac{(t^3+1)-1}{t+1}dt = 6\int\left(t^2-t+1-\frac{1}{t+1}\right)dx$

$= 6\left(\frac{t^3}{3}-\frac{t^2}{2}+t-\log(t+1)\right)+e = 2\sqrt{x}-3\sqrt[3]{4}+6\sqrt[6]{x}-6\log\left(\sqrt[6]{x+1}\right)+e$

$a=2, b=-3, c=6, d=-6$

$\therefore 2a+3b+4c-5d = \ 4-9+24+30 = 49$

14. $\cos^{-1}x-\cos^{-1}\frac{y}{2}=\alpha \Rightarrow A-B=\alpha \Rightarrow \cos(A-B)=\cos\alpha$

$\Rightarrow x\frac{y}{2}+\sqrt{1-x^2}\sqrt{1-\frac{y^2}{4}}=\cos\alpha \Rightarrow \left(\frac{xy}{2}-\cos\alpha\right)^2=\left(1-x^2\right)\left(1-\frac{y^2}{4}\right) \Rightarrow 4x^2-4xy\cos\alpha+y^2=4\sin^2\alpha$

S-II $\sum_{r=0}^{\infty}\tan^{-1}\left(\frac{r+1-r}{1+r(r+1)}\right)=\sum_{r=0}^{\infty}\left(\tan^{-1}(r+1)-\tan^{-1}(r)\right)=\frac{\pi}{2}$

15. $\lim_{x\to 0}\frac{\left(1-\cos\left(\frac{x^3}{3}\right)\right)\left(1-\cos\frac{x^6}{6}\right)}{x^{18}}$

$\lim_{x\to 0}\left(\frac{1-\cos\left(\frac{x^3}{3}\right)}{\left(x^3\right)^2}\right)\left(\frac{1-\cos\left(\frac{x^6}{6}\right)}{\left(x^6\right)^2}\right)=\frac{\left(\frac{1}{3}\right)^2}{2}\times\frac{\left(\frac{1}{6}\right)^2}{2} \Rightarrow \lambda^2=\frac{1}{36\times 36} \Rightarrow \lambda=\frac{1}{36}$

$\therefore 108\lambda = 108\times\frac{1}{36}=3$

16. $\sum_{n=1}^{10}\frac{(n+2)-n}{2\left[n(n+1)(n+2)\right]}=\frac{1}{2}\left[\frac{1}{1.2}-\frac{1}{2.3}+\left(\frac{1}{2.3}-\frac{1}{3.4}\right)+.....+\left(\frac{1}{10.11}-\frac{1}{11.12}\right)\right]=\frac{1}{2}\left[\frac{1}{2}-\frac{1}{132}\right]=\frac{130}{4\times 132}=\frac{65}{264}$

17. No of equivalence relations is 15

18. Mean $\frac{\sum x_i^2}{15}=12 \Rightarrow \sum x_i = 180$ = Correct mean $\mu=\frac{182}{15}$

Correct $\sum x_i^2 = 2295-100+144=2339$

Correct $\sigma^2=\frac{2339}{15}-\mu^2 \quad \therefore 30\left(\sigma^2+\mu^2\right)=4678$

19. Equation of chord is $S_1 = S_{11}$

$\Rightarrow \frac{x\sqrt{2}}{9}+\frac{y}{3}=\frac{2}{9}+\frac{4}{9} \Rightarrow \sqrt{2}x+3y=6 \ \Rightarrow y=\frac{6-\sqrt{2}x}{3}$ .....(1)

Solving ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and line (1) Then we get $x=0, x=2\sqrt{2}$

P is are $A(0,2), B\left(2\sqrt{2},\frac{2}{3}\right) \therefore AB=\frac{2\sqrt{22}}{3} \Rightarrow 3AB=2\sqrt{22}$

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20. $\frac{dy}{dx}=\frac{-3^x\left(3^y-1\right)}{3^y\left(3^x-1\right)}\Rightarrow\int\frac{3^y}{3^{y-1}}dy=\int\frac{3^x}{3^{x-1}}dx\Rightarrow\frac{\log\left(3^y-1\right)}{\log_e^3}=\frac{-\log\left(3^x-1\right)}{\log_e^3}+\frac{\log c}{\log_e^3}$

$\Rightarrow\left(3^x-1\right)\left(3^y-1\right)=C$

$y(1)=1\Rightarrow(3-1)(3-1)=C\Rightarrow C=4\Rightarrow\left(3^x-1\right)\left(3^y-1\right)=4\ \Rightarrow y(2)=8\left(3^y-1\right)=4\Rightarrow 3^y-1=\frac{1}{2}$

$\Rightarrow 3^y=\frac{3}{2}\ \Rightarrow y=\log_3^{3/2}\qquad=1-\log_3^2$

21. Let the critics be $E_1.E_2$ and E_3

Let $P(E_1)$, $P(E_2),P(E_3)$ denotes the probability of the critics E_1,E_2,E_3 to be in favour of the book

Given $P(E_1):P\left(\overline{E_1}\right)=5:2\Rightarrow\ P(E_1),=\frac{5}{7}$ similarly $30\left(\sigma^2+\mu^2\right)=4678\ P(E_3)=\frac{3}{7}$,

The required probability $=\ P\left(E_1\cap E_2\cap\overline{E}_3\right)=\frac{209}{343}$

22. In the expansion $\left(5^{1/2}+7^{1/8}\right)^{1029}$,

$T_{r+1}=1024_{c_r}\left(5^{1/2}\right)^{1024-r}\left(7^{1/8}\right)^r=W24_{c_r}5^{\frac{1024-4}{2}}7^{7/8}.,r=0,1,2,...2024$

If $r=0,8,16....1024$ then T_{r+1} is rational

$\therefore$ The no of terms in $\left(5^{1/2}+7^{1/8}\right)^{1024}+\left(5^{1/2}-7^{1/8}\right)^{1024}$ is 513

$\therefore$ The no of irrational terms = 513-129= 384

23. Let $\overline{a}=(3,8,3)$, $\overline{b}=(3,-1,1)$, $\overline{c}=(-3,-7,6)$, $\overline{d}=(-3,2,4)$

S.D $=\frac{\left|\left(\overline{a}-\overline{c}\ \overline{b}\times\overline{d}\right)\right|}{\left|\overline{b}\times\overline{d}\right|}=\sqrt{270}$

24.

$f(x)$ is non differentiable at $x=\frac{\pi}{4},\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4},\frac{9\pi}{4},\frac{11\pi}{4}$

$\therefore\lambda=6\Rightarrow 5\lambda^2=5=5(36)=180$

25. $\Delta=0\Rightarrow\begin{vmatrix}1&-1&1\\2&2&\lambda_1\\3&-1&4\end{vmatrix}=0\Rightarrow\lambda_1=4$

$\Delta_3=0\Rightarrow\begin{vmatrix}1&-1&5\\2&2&8\\3&-1&\lambda_2\end{vmatrix}=0\Rightarrow\lambda_2=14$

$\therefore\perp r$ Distance from (4,14) to line $3x-4y+4=0$ is $\frac{|3(4)-4(14)+4|}{\sqrt{9}+16}=\frac{50}{5}=10$

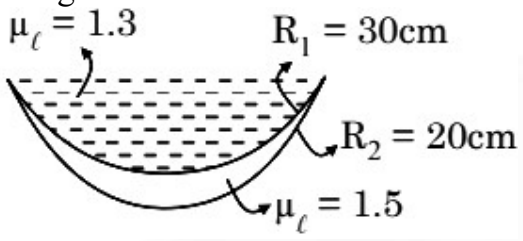
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PHYSICS

26. If both assertion and reason are true but reason is not the correct explanation of assertion.

27. Statement –I is false and statement-II is true

28. Using lens marker's formula



$$\frac{1}{f}=\left(\frac{1.3-1}{1}\right)\left(\frac{1}{\infty}-\frac{1}{(-30)}\right)+\left(\frac{1.5-1}{1}\right)\left(\frac{1}{(-30)}-\frac{1}{(-20)}\right) \quad f=\frac{600}{11}cm$$

29. $$\frac{I_{max}}{I_{min}}=\frac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}-\sqrt{I_2}\right)^2} \quad \frac{I_{max}}{I_{min}}=\frac{\left(\sqrt{4}+\sqrt{1}\right)^2}{\left(\sqrt{4}-\sqrt{1}\right)^2}=\frac{9}{1}$$

By component and dividend

$$\frac{I_{max}+I_{min}}{I_{max}-I_{min}}=\frac{9+1}{9-1}=\frac{10}{8}=\frac{5}{4}= \; x=4$$

30. Statement- I is false statement-II is true

31. The power of an exponent is a number (or) constant

Therefore, dimensionally $\frac{\alpha z}{K\theta}=\left[M^0L^0T^0\right]$

$$\alpha=\frac{K\theta}{z} \qquad \alpha=\frac{\left[M^1L^2T^{-2}\theta^{-1}\right][\theta]}{[L]}=\left[M^1L^1T^{-2}\right]$$

And $P=\frac{\alpha}{\beta}\Rightarrow\beta=\frac{\alpha}{P}$ $\qquad \beta=\frac{\left[M^1L^1T^{-2}\right]}{\left[M^1L^{-1}T^{-2}\right]}=\left[M^0L^2T^0\right]$

32. (A) is true but (R) is false

33. $x=at+bt^2-ct^3$

Velocity, $v=\frac{dx}{dt}=\frac{d}{dt}\left(at+bt^2-ct^3\right)$ $\qquad v=a+2bt-3ct^2$

Acceleration, $a=\frac{dv}{dt}=\frac{d}{dt}\left(a+2bt-3ct^2\right)$ $\quad 0=2b-6Ct \quad t=\frac{b}{3C}$

Then $v=a+2b\left(\frac{b}{3C}\right)-3C\left(\frac{b}{3C}\right)^2 \quad v=a+\frac{b^2}{3C}$

34. For a doped semi conductor in thermal equilibrium

$$n_e n_n=n_i^2 \qquad n_e=\frac{n_i^2}{n_n}=\frac{\left(1.5\times10^{16}\right)^2}{4.5\times10^9 m^{-3}}$$

35. $\in=-L\frac{dI}{dt}$ $\qquad |0.1|=\left|-L\left(\frac{2-(-2)}{0.2}\right)\right|$ L=5mH

36. Horizontal force, F=80 N

$$a_A=a_B=a_C=\frac{F}{5+3+2}=\frac{180}{10}=8\,m/s^2$$

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$5Kg \rightarrow T_1$ $\quad T_1 = 5\times 8 = 40\,N$

$T_1 \leftarrow 3Kg \rightarrow T_2$ $\quad T_2 - T_1 = 3\times 8$

$T_2 - 40 = 24$

$T_2 = 64\,N$

37. Force, $F = YA\dfrac{\Delta l}{l}$

$F = 2\times 10^{11}\times 10^{-4}\left(\dfrac{2l-l}{l}\right)$ $\quad F = 2\times 10^{+7}\,N$

38. Ampere –maxwell law is

$\int \overline{B}.\overline{dl} = u_0 i_c + u_0 \in_0 \dfrac{d\phi_E}{dt}$

Faraday law is $\int \overline{E}.\overline{dl} = \dfrac{d\phi_B}{dt}$

Gauss's law for electricity $\int \overline{E}.\overline{dA} = \dfrac{Q}{\in_0}$

Gauss's law for magnetism $\int \overline{B}.\overline{dA} = O$

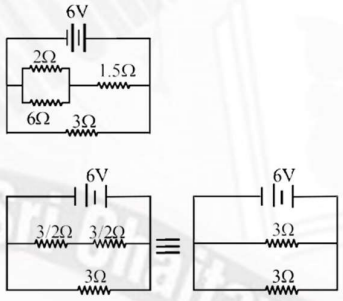
39. Given, $\dfrac{f_1}{f_2} = \dfrac{4}{25}$ RMS speed, $v\alpha \dfrac{1}{\sqrt{f}}$

$\dfrac{v_1}{v_2} = \sqrt{\dfrac{f_2}{f_1}} = \sqrt{\dfrac{25}{4}} = \dfrac{5}{2}$

40. Final momentum

$P = \dfrac{E}{C} = \dfrac{p\times t}{C}$ $\quad P = \dfrac{30\times 10^{-3}\times 100\times 10^{-9}}{3\times 10^8}$ $\quad P = 1\times 10^{-17}\,Kg\,ms^{-1}$

41.



Hence $R_{equ} = \dfrac{3}{2}$

$I = \dfrac{V}{R}$ $\quad I = \dfrac{6}{(3/2)}$ $\quad I = \cancel{6}\times \dfrac{2}{\cancel{3}}$ $\quad I = 4A$

42. Let charge is $q = 10\mu C$ divided into two parts x and $q-x$

From coulomb's law $F = K\dfrac{q_1 q_2}{r^2}$ $\quad F = \dfrac{K\,x\,(q-x)}{r^2}$

For F to be maximum $\dfrac{dF}{dx} = 0 = \dfrac{d}{dx} = \left(\dfrac{kx(q-x)}{r^2}\right) = 0 = x = \dfrac{q}{2}$

Hence charge of the two parts are $5\mu c$ and $5\mu c$

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43. Maximum velocity in SHM, $V_{max}=Aw$

Maximum acceleration in SHM, $a_{max}=Aw^2$

Given, $\frac{a_{max}}{V_{max}}=10$ $\frac{Aw^2}{Aw}=10$ $\rightarrow w=10\,rad/sec$

$x=A\sin\left(wt+\frac{\pi}{4}\right)$ at $t=0\Rightarrow x=5m$ $5=A\sin\frac{\pi}{4}$ $A=5\sqrt{2}\,m$

Maximum acceleration $a_{max}=5\sqrt{2}(10)^2=a_{max}=500\sqrt{2}\ m/s^2$

44. $\frac{Reading-LFP}{UFP-LFP}=const$ $\frac{tp-30}{180-30}=\frac{tq-0}{100-0}$ $\frac{tp-30}{150}=\frac{tq}{100}$

45. Boolean expression of given circuit $Y=A.\overline{B}+\overline{A}.B=A+B$

This is XOR gate

46. Magnetic force acting on electron in a uniform magnetic field is given by

$\overline{F}=q\left(\overline{v}\times\overline{B}\right)$ $5e\hat{K}=e\left[\left(3\hat{i}+5\hat{j}\right)\right]\times\left(Bo\hat{i}+2B_0\hat{j}\right)$

$5e\hat{K}=e\left[6B_0\hat{K}-5B_0\hat{K}\right]$ $B_0=5T$

47. Using ohm's law $R=\frac{V}{I}=\frac{100}{5}=20\Omega$

Impedance is given as $Z=\sqrt{X_2^2+R^2}=\sqrt{3\times400+400}$ $Z=40\Omega$

Power dissipated in circuit is

$P=\left(\frac{V_{rms}}{Z}\right)^2\times R$ $P=\left(\frac{200/\sqrt{2}}{40}\right)^2\times20=250\,W$

48. Given,

Mass of the body , $m=2Kg$

Power delivered by engine, $P=1J/s$

Time, $t=9\,seconds.$

Distance, $x=\sqrt{\frac{2P}{m}}\times\frac{2}{3}t^{3/2}$ $x=\sqrt{\frac{2\times1}{2}}\times\frac{2}{3}(9)^{3/2}$ $x=18\,m$

49. Given, Radius of circular orbit $r=0.5A^0$

Speed of electron, $v=2.2\times10^6 m/s$

Current, $i=\frac{q}{t}=\frac{q}{\left(\frac{2\pi r}{v}\right)}$ $i=\frac{1.6\times10^{-19}\times2.2\times10^6}{2\pi\times0.5\times10^{-10}}$

$i=1.12\,mA$ $i=112\times10^{-2}mA$

50. The moment of inertial of a uniform thin rod about an axis perpendicular axis passing through one end is, $I_1=\frac{ml^2}{3}$ Now, moment of inertia of bent ring

$I_2=\frac{1}{2}mR^2$ $I_2=\frac{1}{2}m\left(\frac{l}{2\pi}\right)^2$ $I_2=\frac{ml}{8\pi^2}$

$\frac{I_1}{I_2}=\frac{\frac{ml_2}{3}}{\frac{ml^2}{8\pi^2}}=\frac{8\pi^2}{3}$ so, $x=8$

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CHEMISTRY

51. $\Delta x = \Delta P$

$$\Delta P^2 = \frac{h}{4\pi} \qquad \Delta v^2 = \frac{1}{2m}\sqrt{\frac{h}{\pi}}$$

52. Statement- I is true statement-II is false

53. Statement- I is true statement-II is false

54. $$\frac{6}{E_m + 30} = \frac{4}{E_m + 16}$$

$$1.5E_m + 24 = E_m + 30$$

55. $$K = \frac{2.303}{t}\log\frac{(a)}{(a-x)}$$

56. $$E^0 = \frac{0.0591}{2}\log Kq$$

$$\log K_q = \frac{2\times 0.22}{0.591}$$

$$\log K_q = 7.44$$

$$K_q = 2.8\times 10^7$$

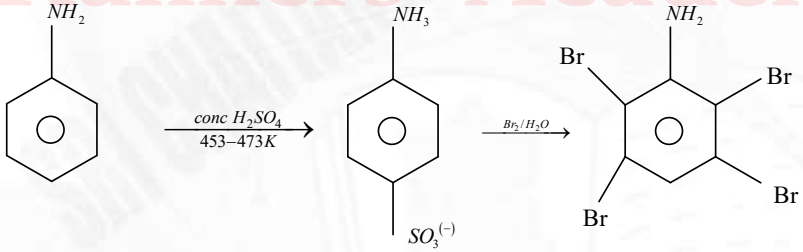
57. (A) is true but (R) is false

58. 4f subsheel – half filled more stable

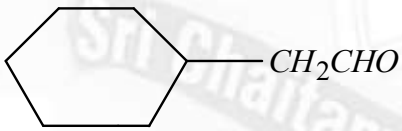
59. EAN= $z - x - 2y = 24 - 3(2\times 6) = 33$

60. $\left[Co(en)_2 Cl_2\right]^+$

61. Ipso substation reaction



62.



63.

$\forall(z=23)$ $n=3$

$Cr(z=24)$ $n=6$

$Mn(z=25)$ $n=5$

64. $P-O$ Bond oder= $\frac{\text{Total number of Bonds}}{\text{Total number of R.S}} = \frac{5}{4} = 1.25$

65. Anomers

66. 2- Ethyl 1,1 di methyl cyclobutane

67. $$C_xH_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$$

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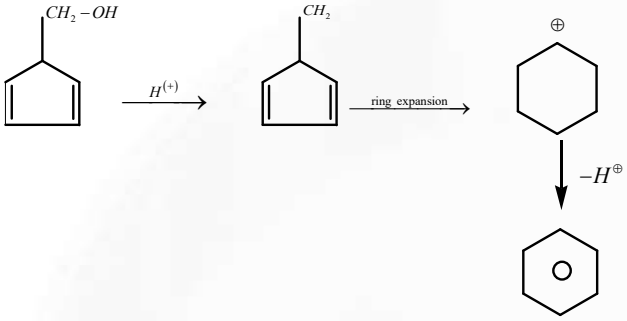
$\left(x+\frac{y}{4}\right)=11$ $\frac{y}{2}=4$

$x+\frac{8}{4}=11$ y=8

$x+2=11$ C_9H_8 $x=9$

68. Dehydroalozination

69.



70. Ammonium ion

71. $r_n=0.529\frac{n^2}{2}=2.116$

$E_n=-\frac{13.6\times z^2}{n^2}=\frac{-13.6\times 1^2}{2^2}=-3.4\,\text{ev}$

$K\in=-E_n=+3.4ev$

72. $2H_2O\rightarrow O_2+4H^++4e^-$

$P^{OH}=12$

73. Two molecules having zero dipole moment

74. 1) C_3O_2 , 2) Fe_3O_4

75. 3 $NaNH_2$

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